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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Machine Learning (course)

Week 8: Assignment 8 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) In AdaBoost, we re-weight points giving points misclassified in previous iterations more weight. Suppose we introduced a limit or cap on the weight that any point can take (for example, say we introduce a restriction that prevents any point's weight from exceeding a value of 10). Which among the following would be an effect of such a modification? (Multiple options may be correct) ***1 point***

- [ ] We may observe the performance of the classifier reduce as the number of stages increase
- [ ] It makes the final classifier robust to outliers
- [ ] It may result in lower overall performance

2) A and B are Boolean random variables. ***1 point***

Given:
$P(A = True) = 0.3, P(A = False) = 0.7, P(B = True|A = True) = 0.4, P(B = False|A = True) =$
$P(B = True|A = False) = 0.6, P(B = False|A = False) = 0.4$. Calculate $P(A = True|B = False)$ by Bayes rule.

- 0.49
- 0.39
- 0.37
- 0.28
- None of the above

3) Which of the following method(s) is not inherently sequential? ***1 point***

- Gradient Boosting
- Committee machines
- AdaBoost

4) In a tournament classifier with N classes. What is the complexity of the number of classifiers we require? ***1 point***

- $\mathcal{O}(N)$
- $\mathcal{O}(N^2)$
-

$\mathcal{O}(N.\,log(N))$

$\mathcal{O}(log(N))$

Check Answers and Submit